



FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 01.08.2026

In addition to part I (the General Handout for all courses, appended to the Time table), this section provides further specific details regarding the course.

Course No : SCM G513
Course Title : Advanced Operations Management
Instructor-in-Charge : AMIT KUMAR JAIN
Instructor(s) : NA
Tutorial/Practical Instructors: NA

1. Course Description:

Types of Manufacturing, Manufacturing, maintenance, quality and service operations management concepts. Manufacturing Management measures, Factory/plant performance, Operations Improvement framework. Maintenance management, optimising maintenance costs; Improving asset lifecycle; Preventive maintenance; Total productive maintenance (TPM); Quality management - practices and processes; Total quality management (TQM); Service Operations Management definition; Customer loyalty and satisfaction.

2. Scope and Objective of the Course:

This course details the various aspects and emerging concepts of operations management as they relate to the manufacturing/production function, maintenance, quality, and service operations. It provides a detailed view of each of the above functions, including their strategic importance, operational aspects, and roles in the supply chain. The objectives of the course are mentioned below:

- This course will give students an overview of the various components of manufacturing and maintenance management as practised on the shop floor. The various kinds of manufacturing operations, the types of jobs performed, and the maintenance activities undertaken to ensure operational readiness and integration with other functions of the industry will be discussed. This will help one understand how the shop-floor execution and upkeep of resources (machines/equipment) are critical to the functioning of a coherent supply chain.
- This will help the students to understand the importance of quality management. Explain the various quality management tools and techniques used in an organisation to ensure that quality products and services are produced.
- This course will provide an overview of the service operations management. Customer service design, service operations and execution, and monitoring service delivery performance will be discussed.

3. Text Books:

T1: Roberta S. Russell, Bernard W. Taylor, Venkataramanaiah Saddikuti, Pavan Kumar Gudavalleti; *Operations and Supply Chain Management*, 10ed (An Indian Adaptation); Wiley, 2023, ISBN: 9789354644092.

T2: F. Robert Jacobs, Ravi Shankar, Richard B. Chase, *Operations and Supply Chain Management*, 17th Ed, McGraw-Hill, 2023, ISBN-10: 9355325223.

T3: Sanjeev K. Bordoloi, James A. Fitzsimmons, Mona J. Fitzsimmons; *Service Management*, 9th ed.; McGraw-Hill India, 2023, ISBN: 9789355324870.



T4: S.K. Srivastava: *Maintenance Engineering (Principles, Practices and Management)*, S. Chand & Co., New Delhi, 1998, ISBN: 9788121926447

4. Reference Books:

R1: Jay Heizer, Barry Render, Chuck Munson; *Operations Management: Sustainability and Supply Chain Management*, 14ed; Pearson, 2023, ISBN: 9780137649136.

R2: Lindley R. Higgins and R. Keith Mobley: *Maintenance Engineering Handbook* - 7th ed. McGraw-Hill, 2002.

R3: Douglas C. Montgomery; *Statistical Quality Control: A Modern Introduction*, 6ed; Wiley, 2010, ISBN: 9788126525065.

R4: Gerard Cachon, Christian Terwiesch; *Matching Supply with Demand: An Introduction to Operations Management*, 4th ed.; McGraw-Hill, 2023, ISBN: 9789355325051.

R5: Tina Kanti Agustiady, Elizabeth A. Cudney; *Total Productive Maintenance: Strategies and Implementation Guide*; CRC Press, 2016, ISBN: 9781482255409.

R6: Selected journal papers shared during lectures.

5. Course Plan:

Week No. Lecture Session	Module	Topics	Ref.	Learning outcomes
1 L: 1-3	Manufacturing Systems	Course introduction. Definition and scope of operations management. Role of operations management in organisations. Functions and challenges of operations management. Types of manufacturing (job, batch, mass, continuous). Product-process matrix: linking volume-variety with process choice. Breakeven Analysis.	T1, Ch 1 and 6 Lecture Notes	Understand the fundamentals of operations management and explain how different manufacturing systems and process choices support organisational operations.
2 L: 4-6	Manufacturing Management Measures	Productivity measures (labour, capital, multi-factor productivity). Numerical: productivity index calculation. Cycle time, throughput, takt time. Numerical: takt time and throughput calculation for a given demand/production scenario. Capacity utilisation and efficiency measures. Numerical: capacity utilisation and efficiency exercise from plant output data.	T1, Ch 1, 7 and 15 Lecture Notes	Calculate and interpret key operational performance measures, including productivity, cycle time, throughput, takt time, capacity utilisation, and efficiency.
3 L: 7-9	Fundamentals of Maintenance Management	Role of maintenance in operations management. Maintenance definition, need, objectives, functions, benefits, and impact. Standards in maintenance management.	T4, Ch 1 Lecture Notes	Understand the role, objectives, functions, benefits, and standards of maintenance management in



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				supporting effective operations.
4 L: 10-12	Maintenance Strategies	Evolution of maintenance. Types of strategies: reactive, preventive, predictive, and prescriptive maintenance. The bathtub curve and product failure behaviour; predicting failures with condition monitoring; understanding the P-F interval and the P-F curve; selecting a maintenance strategy based on machine asset type; functional blocks of digital maintenance; and sustainability aspects.	T4, Ch 5 Lecture Notes	Understand the evolution of maintenance, compare maintenance strategies, and explain failure behaviour, condition monitoring, digital maintenance, and sustainability concepts.
5 L: 13-15	Optimising Maintenance Costs	Measuring failures and KPIs: failure rate, MTTF, MTBF, MTTR, downtime, availability, maintainability. Maintenance trade-offs: preventive cost vs. breakdown cost, optimal maintenance interval. Preventive Maintenance (PM) modelling. Hands-on exercise for designing PM schedules.	T4, Ch 4 and 14 Lecture Notes	Calculate maintenance performance measures and develop preventive maintenance schedules to improve equipment reliability and availability.
6 L: 16-18	Understanding Predictive Maintenance	Predictive maintenance workflow: acquire data, pre-process data, identify health indicators, train the model using machine learning, deploy, and integrate. Remaining useful life, hyperparameter tuning, performance assessment. Industrial use cases and success stories. Hands-on: building a predictive maintenance model with a linear regression algorithm.	R2, Ch 3 Lecture Notes	Understand the predictive maintenance workflow and develop a basic machine learning model for predicting equipment condition and remaining useful life.
7 L: 19-21	Total Productive Maintenance (TPM)	TPM philosophy, objective of TPM, TPM vs traditional planned maintenance, 5S foundation, TPM pillars (autonomous maintenance, planned maintenance, quality maintenance, focused improvement, early equipment management, training and education, safety, health, environment, TPM in administration), benefits of TPM. Factory/plant performance: Overall Equipment Effectiveness (OEE), productivity loss: understanding the six big losses, world-class OEE. Case-based calculation and discussion.	T4, Ch 10 and 13 R5 Lecture Notes	Understand the principles of Total Productive Maintenance (TPM) and evaluate equipment performance using Overall Equipment Effectiveness (OEE).
8 L: 22-24	Improving Asset Lifecycle	Asset lifecycle stages: acquisition, operation, maintenance, disposal/replacement. Life-cycle costing: total cost of ownership across the asset's life and replacement problems.	Lecture Notes	Understand asset lifecycle stages and apply life-cycle costing concepts to support asset replacement and management decisions.
9 L: 25-27	Quality Management	Definition, need. Quality management system. Quality in service. Cost of quality: prevention costs, appraisal costs, internal failure costs, external failure costs. Core	T1, Ch 2	Understand quality management principles, quality systems, quality costs,



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		quality practices: standardisation, process control, quality audits, corrective/preventive action. Quality awards. Quality certifications and standards.	Lecture Notes	and key practices for improving product and service quality.
10 L: 28-30	Quality Tools	The 7 QC tools (Pareto chart, cause-and-effect/fishbone diagram, histogram, check sheets, scatter diagrams, control charts, flowcharts). Root cause analysis techniques: 5 whys, fishbone diagram - applied exercise.	T1, Ch 2 Lecture Notes	Apply quality control tools and root cause analysis techniques to identify, analyse, and solve quality problems.
11 L: 31-33	Total Quality Management (TQM)	TQM philosophy: customer focus, continuous improvement, employee involvement, process-centred thinking. Key contributors. Operations improvement framework: PDCA (plan-do-check-act) cycle and kaizen in practice. Six sigma basics: DMAIC framework and its relationship to TQM.	T1, Ch 2 Lecture Notes	Understand Total Quality Management (TQM) principles and apply continuous improvement approaches, including PDCA, Kaizen, and Six Sigma.
12 L: 34-36	Statistical Process Control	Why statistical thinking matters in quality: variation - common cause vs. special cause. Concept of a process "in control" vs. "out of control". Control charts for variables: x-bar and R charts, construction and interpretation. Numerical: build an x-bar and R chart from sample data, compute control limits. Control charts for attributes: p-chart, np-chart, c-chart, u-chart, when to use which. Numerical: attributes chart construction from data.	T1, Ch 3 R3 Lecture Notes	Apply statistical process control techniques to monitor process variation and construct and interpret control charts for quality improvement.
13 L: 37-39	Process Capability Analysis	C_p , C_{pk} : measuring whether a process meets specification limits, numerical: C_p/C_{pk} calculation and interpretation (capable vs. not capable). Interpreting out-of-control signals: run rules, patterns, trends. Applied exercise: given a dataset, identify whether a process is in control and capable, and recommend corrective action.	T1, Ch 3 Lecture Notes	Evaluate process capability and control using C_p , C_{pk} , and control chart analysis to identify quality issues and recommend corrective actions.
14 L: 40-42	Service Operations Management	Understanding services. Service classification frameworks: service process matrix (degree of labour intensity vs. customisation/interaction). Service quality dimensions: (reliability, responsiveness, assurance, empathy, tangibles). Managing service operations: capacity and demand management, globalisation of services. Customer loyalty and satisfaction.	T3, Ch 1, 6, 10, and 11 Lecture Notes	Understand service operations concepts and evaluate service quality, capacity management, and customer satisfaction in service organisations.



6. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time*	Nature of component Close Book (CB) / Open Book (OB)
Mid-Semester Test	90 min	25	05/10 AN2	CB
Quiz	30 min	10	To be announced in class	CB, Best 2 of 3
Project	-	30	To be announced in class	OB
Comprehensive Examination	3 h	35	03/12 AN	CB + OB
Total		100		

*Date and time are tentative and subject to change. Stay updated with the information from the exam office.

7. Chamber Consultation Hour: Chamber no 2219-A; Wednesday, 5.00 PM to 6.00 PM

8. Notices: All the notices regarding the course will be emailed or displayed on the Nalanda or will be displayed on the Mechanical Engineering Department Notice Board only.

9. Make-up Policy: Make-up will be permitted only for genuine cases with prior permission. For the Quiz and Project, no make-ups will be granted.

10. Attendance Policy: Each student is expected to attend all classes and participate in all evaluation components.

11. Minimum passing Marks criteria: Failure to appear for the mid-semester/comprehensive exam. Scoring less than 20 marks out of 100 or a suitable modification at the discretion of the IC, with the concurrence of the Department.

12. Note (if any): NA

Instructor-in-charge

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